

DSA210 - Project Proposal

Football Analytics

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Motivation

Predicting soccer match results with a great degree of confidence has numerous potential financial gains by selling the data to betting/analytics companies or betting on matches where the model has an edge over the bookmaker, eventually turning in a profit. However, most in-house models are trained on coarse-grained data such as end of match statistics, lineups or news articles. These efforts perform well enough for low confidence interval match predictions and as a proof of concept. But they underperform against a betting market, since bookmakers use the same data to set the odds. This is due to the hard-to-obtain nature of fine-grained data. Sports data collection companies provide this data in a limited manner, sometimes with outrageous pricing. Thus, a model that is trained on multiple aspects of soccer data, along with fine-grained data, has the potential to improve predictive performance relative to bookmaker odds.

Fortunately, most betting platforms provide the option to bet on live matches. Live matches introduce novel statistics during gameplay that cannot be taken into account before the match starts. A player may underperform greatly, weather conditions may have adverse effects on team performances or the strategy of the teams may collide in an unpredictable way. This provides a setting where models that incorporate fine-grained and real-time data can be evaluated more effectively.

Objective

The goal of this project is to predict match outcome probabilities using both pre-match and in-game event data, and to evaluate whether fine-grained event data improves predictive performance compared to models relying only on coarse-grained features and bookmaker implied probabilities.

Data Collection

What I did until now

I have successfully reverse engineered a public facing product of SportRadar, a company that provides fine-grained match data to analytics companies. Via the help of their API and the reverse engineered endpoints, I have collected over 70,000 matches, with 10,000 containing the most fine-grained event data.

What I will do

I will keep on collecting more data to enrich the dataset. I plan on including: expected goals values, per player statistics, pre-match betting odds, historical features like momentum, and other real-time coarse-grained data that are currently missing. I will also examine the distribution of leagues, seasons and match coverage to identify potential imbalances in the dataset.

Data Analysis and Preparation

I will uncover potential biases in my dataset by training a simple random forest model and extracting feature importance with SHAP. I will specifically look for: league bias, team strength imbalance, temporal drift and missing data patterns. Additionally, I will perform exploratory data analysis to better understand the structure of event sequences and feature distributions.

Model Training

I will train models to predict match outcome probabilities. The input will consist of both static pre-match features and sequential in-game event data such as event type, timestamp, player information and expected goals.

I will be training a transformer model to capture the sequential nature of the event data. The sequence will represent ordered match events, and the model will output probabilities for match outcomes (win/draw/loss) at different time steps.

As a baseline, I will also use XGBoost by converting the sequential data into aggregated features. This will provide a comparison between sequence-based models and traditional tabular approaches.

Evaluation

The models will be evaluated using log loss and Brier score to measure the quality of predicted probabilities. I will use a time-based train-test split to avoid data leakage, training on past matches and testing on future matches.

Additionally, I will compare model predictions against bookmaker implied probabilities to evaluate whether fine-grained data provides any measurable improvement.